

Computational Fluency Short Course

Introduction

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<https://github.com/brownridd/cfsc26>

Expectations

The short course examines *frameworks* and *processes* for **research computation**.
The focus is a mix of career development and technical skill.

Plan to develop *coding* and detailed methods in your individual labs.

You will already know some of the things, probably not all the things, and maybe not systematically organized.

Don't be afraid to be wrong. Ask for help when you want it. Help others when you can (*if they want you to!*). *Be professional and kind.*

I have my ways. Develop any process that works for you (and your colleagues...).

You will need to learn things your mentors don't know, because the practice of science is changing faster than the people doing it.

Logistics

Schedule:

today 2-4, Thurs 2-5, Fri 9-12

Mon 9-12, Tues 9-12, Tues 2-5

Wed 9-12

Intended topics:

Computer architecture

User Interfaces

Filesystems

Software dependency

Version control

Coding patterns

Resource management

Troubleshooting

Using AI

Project best practices

There is room for improvisation: some of what we cover will adapt to timing and your questions.

Course materials will accumulate at

<https://github.com/brownridd/cfsc26>

Past innovations decode present complexity



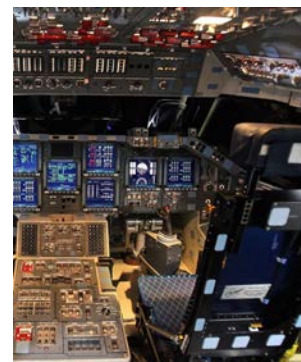
1930s



1960s



1980s-1990s



<https://en.wikipedia.org/wiki/File:DH87.jpg>; [Hornet_moth_dh87b_g-adne_arp.jpg](https://en.wikipedia.org/wiki/File:Hornet_moth_dh87b_g-adne_arp.jpg);
[British_Airways_Trident3B_\(7107744185\).jpg](https://en.wikipedia.org/wiki/File:British_Airways_Trident3B_(7107744185).jpg); [TridentFlightDeck.JPG](https://en.wikipedia.org/wiki/File:TridentFlightDeck.JPG); [STS-121-DiscoveryEnhanced.jpg](https://en.wikipedia.org/wiki/File:STS-121-DiscoveryEnhanced.jpg)
https://commons.wikimedia.org/wiki/File:Space_Shuttle_Endavour%27s_Control_Panels.jpg

What is the purpose of programming for research?

For most scientific applications, code is not "the product", it is a means to an end. Code is an exact, concrete manifestation of the quantitative logic of the study. The code matters only to the extent the study is informative.

Scientific code should be (at minimum):

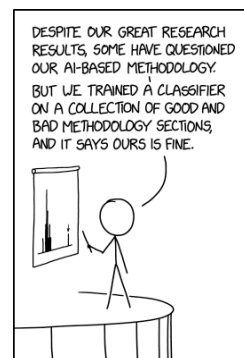
- *Useful*: it accomplishes some scientific goal
- *Honest*: it actually does the thing it claims to do
- *Reproducible*: others can run it and get exactly the same results
- *Auditable*: others can inspect and understand what it does

It is your responsibility to produce code that meets these criteria, even when using AI, online sources, or any other process.

*The **precise specification** of the problem, your **choice** of methods, and **validation** of results is the real work (and the hard part); the implementation is just details.*

AI services raise new and wide ranging ethical considerations

- Copyright
- Data risk
 - Exposure of data to AI providers and third parties
 - Corruption or loss of your own data
- Environmental and social harm
- Cognitive impairments and undetected bias
- Unstable and unpredictably revised AI performance



<https://xkcd.com/2451>

Scientific use raises additional issues for **rigor**, **reproducibility**, and **integrity**.

You maintain final responsibility for your decisions of what to use and for all outcomes.